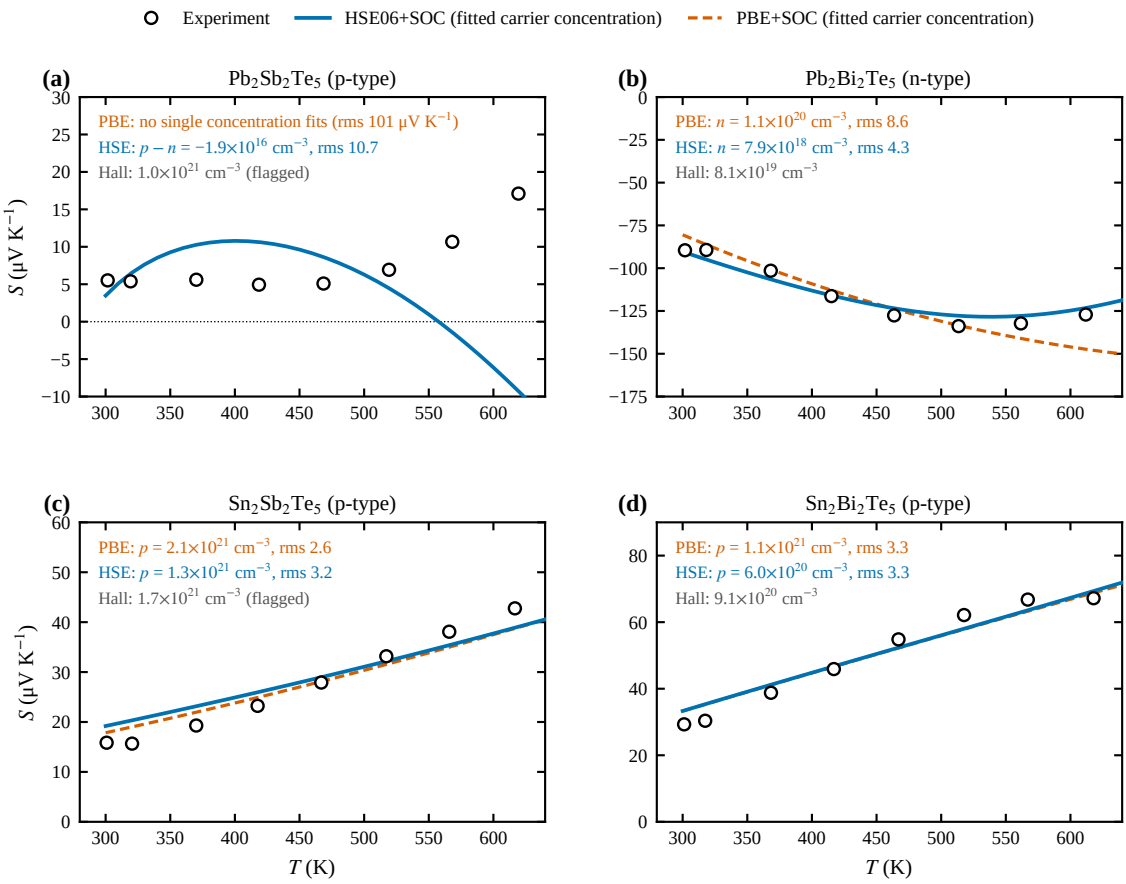
Experiment
HSE06+SOC (fitted carrier concentration)
PBE+SOC (fitted carrier concentration)
(a)
Pb2Sb2Te5 (p-type)
PBE: no single concentration fits (rms 101 μV K−1)
HSE: p − n = −1.9×10^16 cm−3, rms 10.7
Hall: 1.0×10^21 cm−3 (flagged)
(b)
Pb2Bi2Te5 (n-type)
PBE: n = 1.1×10^20 cm−3, rms 8.6
HSE: n = 7.9×10^18 cm−3, rms 4.3
Hall: 8.1×10^19 cm−3
(c)
Sn2Sb2Te5 (p-type)
PBE: p = 2.1×10^21 cm−3, rms 2.6
HSE: p = 1.3×10^21 cm−3, rms 3.2
Hall: 1.7×10^21 cm−3 (flagged)
(d)
Sn2Bi2Te5 (p-type)
PBE: p = 1.1×10^21 cm−3, rms 3.3
HSE: p = 6.0×10^20 cm−3, rms 3.3
Hall: 9.1×10^20 cm−3
S (μV K−1)
T (K)